

PQC-Based Secure Access Manager for IJTAG Network

Prajwal G S Basavaraj¹
*Dept. of Electronics and
 Communication Engineering,
 PES University
 Bengaluru, India*
prajwalgsbasavaraj@gmail.com

Abhinav S²
Dept. of Electronics and
Communication Engineering,
PES University
Bengaluru, India
abhinavsasivala02@gmail.com

Arun Kumar N³
Dept. of Electronics and
Communication Engineering,
PES University
Bengaluru, India
arunbnn007@gmail.com

Sudeendra Kumar K⁴
Dept. of Electronics and
Communication Engineering,
PES University
Bengaluru, India
sudeendrakumark@pes.edu

Abstract—As the complexity of System-on-Chip (SoC) designs is increasing, the number of instruments embedded within the chip has also grown significantly. On the other hand, the flexibility provided by the IEEE 1687 standard has led to security issues such as unauthorized access, leakage of information contained in the scan chain, replay attacks, and misuse of the test interface. The existing security schemes for JTAG are primarily based on conventional cryptographic techniques like RSA and lightweight ciphers. As these techniques are vulnerable and it is safe to migrate to post quantum ciphers even for the chip test security for the devices which have long utilization life especially in the automobiles and industrial control. This work addresses the security issues by developing a secure JTAG access architecture based on lattice-based Post Quantum Cryptographic (PQC) techniques. The proposed solution is a combination of ML-KEM and ML-DSA for a secure JTAG. ML-KEM is used for secure session key establishment. ML-DSA is used for authentication of test access requests and unlocking the Access Manager. In comparison to existing schemes, based on classical cryptographic schemes, the authorization time/access time of the proposed PQC based mechanism is high. The security analysis shows that, the proposed scheme is a countermeasure against brute-force and scan sniffing attacks on par with the existing solutions.

Index Terms—IJTAG, IEEE 1687, Post-Quantum Cryptography, ML-KEM, ML-DSA, Access Manager, SHAKE, SHA3, NTT.

I. INTRODUCTION

With the rapid progress of integrated circuit technology, current System-on-Chip (SoC) design has reached unprecedented levels of complexity in integrating large arrays of embedded instruments for in-field monitoring and performance evaluation. With regard to facilitating access to on-chip instruments, the IEEE 1687 (JTAG) standard has become the preferred choice for structured scan networks, facilitating flexible and dynamic access through Segment Insertion Bits (SIBs) [1], as shown in Fig. 1. JTAG has now become an integral component of Silicon Lifecycle Management (SLM), which requires monitoring, maintaining, and optimizing SoC devices in real time. However, the flexibility of the JTAG network also brings forth critical security risks. In an unsecured IEEE 1687 system, any malicious entity with access to the Test Access Port (TAP) of the system has the potential to traverse through the scan chain, access embedded instruments, and steal critical on-chip data [2]. This security risk is present

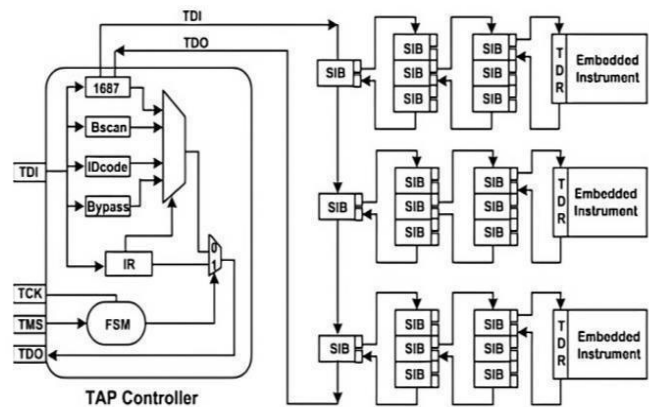


Fig. 1. Basic Structure of IJTAG

throughout the system lifecycle, including production testing, post-silicon validation, embedded software development, and infield diagnostics. An attacker has access to various attack vectors through the JTAG network, including data sniffing attacks, denial of service attacks, SAT-based structural analysis attacks, replay attacks, and brute-force guessing attacks, which may result in compromising the intellectual property and functionality of the SoC.

Significant research efforts have already focused on these issues. Dworak et al. [3] proposed Locking SIBs (LSIBs) and limited access to the scan segment with the aid of hardwired n -bit keys. Gupta et al. [4] extended the LSIB concept to parallel architectures. Liu and Agrawal [5] proposed a variant of LSIBs and added Secure LFSRs for data randomization in the locked state. Zygmuntowicz et al. [6] proposed Honeypot LSIBs, and Kan et al. [7] proposed post fabrication stub chain programming for anti-reverse engineering. Authentication-based solutions comprise the Secure Scan Chain (SSC), a multi-level secure JTAG architecture proposed by Pierce and Tragoudas [8], Clark's key-based instruction set mapping [9], Trivium stream cipher encryption proposed by Rosenfeld and Karri [10], block cipher-based protection proposed by Da Silva et al. [11], and PUF-LFSR-based key gating for each instrument proposed by Kumar et al. [12].

Recently, Ithal et al. [13] proposed a variant combining

TABLE I
PREVIOUS WORKS ON JTAG SECURITY

Work	Threat Model	Countermeasure	Limitation	Quantum-Safe
Dworak [3]	Scan Snooping	Locking SIBs (n -bit key)	Static Key, brute-forceable	No
Rosenfeld [10]	Scan sniff, replay	Trivium stream cipher	Classical stream cipher	No
Liu [5]	Scan Snooping	LFSR key gating	Classical LFSR, predictable	No
Kumar [12]	Reverse engineering	PUF-LFSR per instrument	PUF aging; classical only	No
Ithal [13]	Scan sniff, replay	SHAKE-256 + Grain-128	Classical stream cipher	No
Katti [14]	Scan sniff, replay	OTP + Trivium + AM	Classical crypto only	No
Proposed work	Quantum + HNDL	ML-KEM + ML-DSA (FIPS 203/FIPS 204)	High Hardware Overhead	Yes

SHAKE-256 on the TMS port and Grain-128-based encryption instrument output data. Katti et al. [14] proposed a Central Access Manager (AM) for a multistage OTP-based unlock protocol, instrument clustering, selective Trivium-based output encryption, and TRNG-based masking of TDO. However, all the proposed IJTAG security solutions rely solely on classical cryptographic primitives, which are all vulnerable to attacks by a quantum computer. As we know, Shor's algorithm can solve RSA and ECC, which are the bases of key exchange and authentication in most of today's secure test solutions.

In addition, Grover's algorithm reduces the effective length of symmetric-key algorithms by half. In fact, SoC in automotive, aerospace, and industrial applications can be expected to operate for ten or more years, and thus the issue of quantum attacks is not in the future but here today. Chips used in automotive, avionics, and military applications will have a shelf life of 12–18 years. The current security schemes employed for IJTAG security will be vulnerable in chips with longer shelf life. It is highly necessary to adopt PQC schemes even for testing and debugging security of chips in that direction. This is the first endeavor of employing lattice-based PQC for IJTAG security. To address this gap, this paper proposes a post-quantum cryptography (PQC)-based security framework for IJTAG-enabled SoCs, utilizing two NIST-standardized algorithms: the Module Lattice-Based Key Encapsulation Mechanism (ML-KEM, FIPS 203) [15] for a secure session key establishment, the Module Lattice-Based Digital Signature Algorithm (ML-DSA, FIPS 204) [16] for authenticating test commands and unlocking the Access Manager, and a SHAKE-256 [17] stream cipher for selective instrument output encryption. These algorithms are integrated into an improved Access Manager (AM) that maintains the multi-stage OTP and ML-DSA-based unlock protocol, instrument clustering, selective SHAKE-256 stream cipher-based instrument output encryption, and TRNG-based TDO masking, replacing the classical key exchange with quantum-resistant alternatives. The proposed framework offers:

- ML-KEM-based session key encapsulation during provisioning, ensuring session keys remain secure against quantum adversaries,
- ML-DSA-based digital signatures for authenticating every test access request made by the tester,
- FSM-based AM ensuring strict sequential unlock using

OTP-stored ML-DSA verification keys, with all outputs masked by TRNG until authentication is complete,

- Selective SHAKE-256 stream cipher-based encryption of high-priority instrument outputs post-unlock, 80-bit key from ML-KEM and per-session IV from TRNG.

This paper is organized as follows. The next section discusses previous approaches to mitigate the security threats and explains the threat models. Section III discusses the methodology proposed. Section IV presents the implementation and results.

II. BACKGROUND AND THREAT MODEL

A. Background

The IJTAG standard, or IEEE 1687, provides a reconfigurable scan network (RSN) that is rooted in the JTAG TAP controller [1], [2]. Instruments are selectively enabled by writing logic 1 to their associated SIBs, thereby dynamically changing the length of the scan path. This is a key aspect of the IJTAG standard for SLM but is a well-studied threat model [3], [13]. Prior IJTAG security schemes and their threat models, countermeasures, and limitations are summarized in Table I, where we can see that all prior work shares a common limitation: their sole use of classical cryptography. Notably, the post-quantum standardization of ML-KEM [15] and ML-DSA [16] algorithms by NIST is now complete. Thus, it is timely and necessary to re-examine the security of IJTAG in a threat model that is quantum-capable.

B. Threat Model

To the Access Manager model [14] we incorporate a quantum-capable adversary. Two adversaries are considered: one is a Harvest-Now-Decrypt-Later (HNDL) adversary, who gathers ciphertexts during the provisioning phase and decrypts them after a quantum computer is acquired, and the second is a quantum adversary, who uses quantum computers in real-time to break classical public key cryptography schemes during the test session. Both adversaries have complete physical access to the TAP interface and knowledge of the design architecture, but do not have access to OTP internal information or the TRNG hardware root of trust. Under such a framework, classical key exchange methods (such as RSA encryption of device keys) are all vulnerable. Classical digital signatures of test commands are also vulnerable to forgery by a quantum attacker. In the

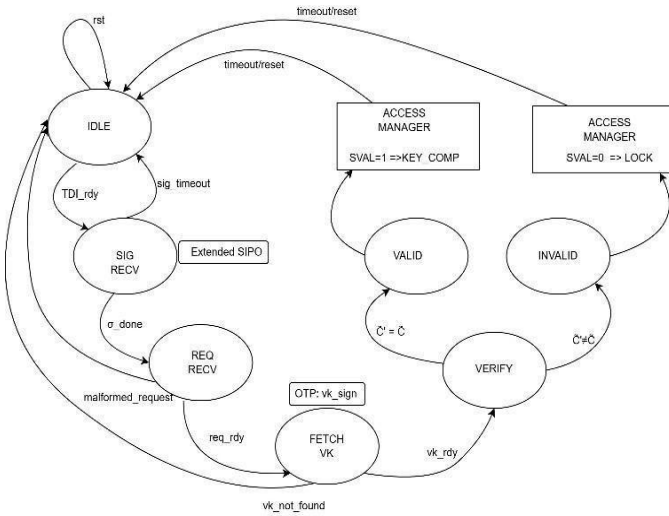


Fig. 3. FSM of ML-DSA

ication Block computes ML-DSA.Verify(vk_sign , request, σ). Failure of the verification immediately leads to LOCK, with TDO driven by J_TRNG (junk TRNG) noise. Only after successful verification does the AM proceed with the pipeline of key, instruction, and data comparison. This guarantees that even an attacker aware of the FSM topology and having observed all valid unlock sequences is unable to construct a new valid TDI frame without possession of the ML-DSA signing key, held exclusively only by the authorized test authority and never transmitted over the IJTAG interface. It is noted that the ML-DSA is used exclusively for authentication and integrity verification of test commands, not for encryption. The encryption of instrument output is handled separately by SHAKE-256 stream cipher. The ML-DSA FSM is shown in Fig. 3.

D. Enhanced Access Manager FSM

The Access Manager (AM) is realized as a secure finite state machine (FSM) having six states: LOCK, SIG_VERIFY, KEY_COMP, INSTN_COMP, DATA_COMP, and UNLOCK, as depicted in Fig. 4.

The finite state machine starts from the LOCK state when the reset signal is applied. An extended SIPO register is employed to capture the incoming TDI stream. The extended SIPO includes the ML-DSA signature field.

FSM state transitions:

- From LOCK, upon receiving a valid TDI frame, the FSM moves to SIG_VERIFY, where ML-DSA verification of the request packet takes place in the ML-DSA verification block.
- Failure of verification causes SVAL = 0 and returns the FSM to LOCK, where TDO is driven by J_TRNG. Upon successful verification, i.e., SVAL = 1, the FSM moves to KEY_COMP, where the 32-bit key is compared with the value stored in OTP.
- If they match, then LVAL is generated and, on first unlock, key_state = 1 is irrevocably set in OTP. The FSM then moves to INSTN_COMP, where the 32-bit instruction word is decoded and sets the 2-bit cluster select

line CLUS_SEL driving DEMUX to select the target MSIB.

- If decoding is valid, then IVAL = 1 and the FSM moves to DATA_COMP, where the 31-bit data word is compared with cluster-specific expected values. If they match, then DVAL = 1 and the FSM moves to UNLOCK, allowing scan data to route to the target cluster and de-asserting J_TRNG.
- If KEY_COMP, INSTN_COMP, and DATA_COMP do not match, then the FSM returns to LOCK. The MUX_SEL selects TDI through a single MSIB during cluster-specific modes and through all MSIBs during production and evaluation modes.

E. Selective Output Encryption and TDO Masking

Output protection employs the two-layer approach. When the FSM is in any state other than UNLOCK, the output MUX feeds pseudo-random information from J_TRNG into TDO, hiding the true state of the instrument and making SAT-based structural analysis via SAT attack algorithms impossible. Once in the UNLOCK state, the outputs of the instruments are passed through a proprietary decoder that uses per-instrument DONE signals to determine eligibility for encryption. The outputs of high-priority instruments are encrypted by SHAKE-256 stream cipher, utilizing an 80-bit key derived from the shared secret K determined by the ML-KEM and a new 80-bit IV provided by TRNG for each unlock session, thus preventing SHAKE-256 stream cipher output reuse from one session to another. Low-priority outputs are not encrypted and are made available on TDO only upon completion of the AM unlock protocol. This approach optimizes test time overhead by only applying the 1600-cycle SHAKE-256 stream cipher initialization overhead to high-priority outputs.

F. Replay and Side-Channel Attack Resistance

The two-phase OTP key protocol is used, where cluster keys are used in the first reset (key_state = 0) and the chip-specific TRNG-derived key is used in subsequent resets (key_state = 1) to prevent replaying of captured unlock sequences. The binding of each request to its corresponding cluster, instruction, and data in ML-DSA signatures prevents partial replay attacks where an adversary attempts to replace some of the TDI

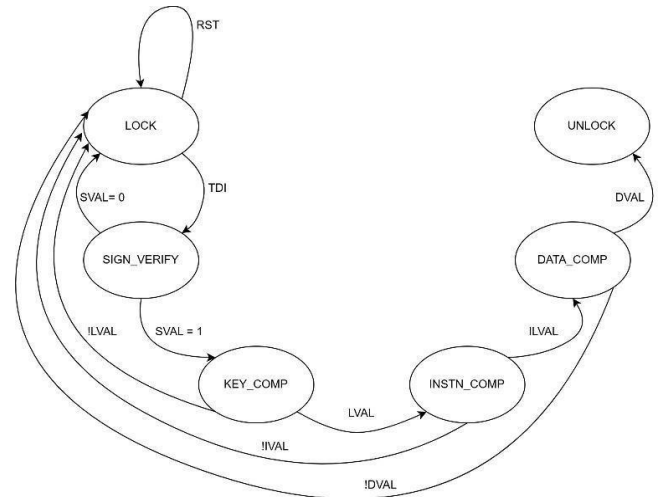


Fig. 4. FSM of the Access Manager

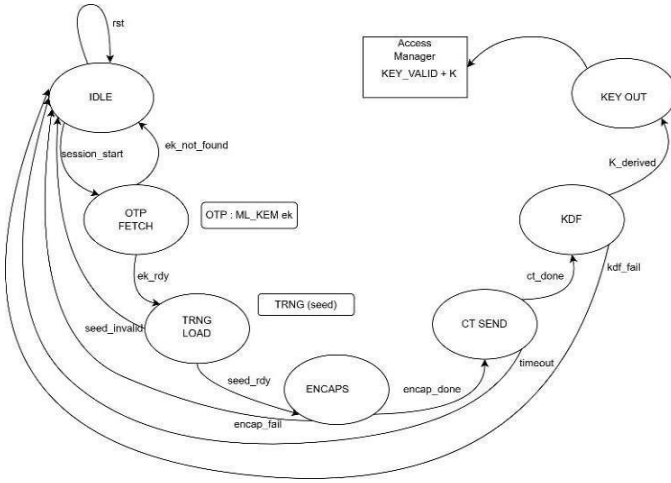


Fig. 5. FSM of ML-KEM

values using a captured signature. For side-channel resistance, J_TRNG continuously generates statistically independent noise on TDO during locked states, ensuring that the Hamming distance of TDO does not leak any correlation to the FSM state and values within the instrument. The ML-KEM encapsulation and ML-DSA verification are done in hardware blocks and therefore do not leak timing side channels at the scan interface level. Resisting PSCA through algorithmic-level countermeasures is considered a complementary topic and is beyond the scope of this work.

IV. IMPLEMENTATION AND RESULTS

The proposed PQC-based secure access manager was implemented in synthesizable Verilog and simulated in Cadence Xcelium simulator and synthesized with the SAED90nm library in Cadence Genus.

A. Hardware Module Organization

The ML-DSA core is built around a single Keccak- f [1600] permutation inside a unified SHAKE engine that supports SHAKE-128 and SHAKE-256 modes through runtime channel multiplexing, with a single Keccak instance serving both channels sequentially via save-buffer. The implementation follows the MLDSA-65 parameter set of FIPS 204 [16] ($K = 6$, $L = 5$, $\lambda = 192$), offering 192-bit post-quantum security with a signature size of 3,309 bytes and a verification key of 1,952 bytes.

Number theoretic transforms (NTT and INTT) are implemented as iterative in-place operations over the MLDSA-65 ring ($q = 8,380,417$) in the Montgomery domain, using a 4-stage pipelined Montgomery multiplier and a radix-2 butterfly unit.

The verification is handled by a 16-state FSM that sequentially handles signature unpacking, norm checking, ExpandA matrix generation, matrix-vector multiplication, hint application, and final challenge comparison.

The ML-KEM core follows the MLKEM-768 parameter set of FIPS 203 [15], generating a ciphertext CT of 1,088 bytes and a 32-byte shared secret K per session. A single

Keccak- f [1600] instance, configurable at runtime to support SHA3-256, SHA3-512, SHAKE-128 and SHAKE-256, forms the hash engine of the ML-KEM core. At the start of each test session, the chip reads the public encapsulation key ek from OTP, obtains fresh randomness from the on-chip TRNG, and runs $ML\text{-}KEM.\text{Encaps}(ek)$ to produce ciphertext CT and shared secret K . The CT is shifted out to the design house via TDO, while K is retained on-chip and passed to the KDF (Key Derivation Function). The incoming TDI stream is captured by an Extended SIPO register of 26,567 bits, which holds the 26,472-bit MLDSA-65 signature σ along with the 95-bit OTP unlock frame consisting of the 32-bit unlock key, 32-bit instruction word, and 31-bit data word.

High-priority instrument outputs are encrypted using a SHAKE-256 stream cipher, keyed with an 80-bit session key derived from the ML-KEM shared secret K through the KDF block, and a fresh 80-bit IV generated by the TRNG for each unlock session. This encryption block reuses the same shared Keccak- f [1600] core, since MLDSA, MLKEM, and the SHAKE-256 stream cipher encryption operations are mutually exclusive in time.

The most significant area saving in the design comes from this Keccak sharing strategy. Since ML-KEM encapsulation and ML-DSA verification are never simultaneous, a single shared Keccak- f [1600] instance serves both, saving approximately 0.166 mm² area. This forms the critical path of the design, limiting the operating frequency to 100 MHz.

B. Resource Utilization

The proposed design was synthesized using Cadence Genus with the SAED 90nm standard cell library at 100 MHz. The area and power results are summarized in Table II. Among the two cryptographic cores, the ML-DSA verification core occupies the most area at 4.192 mm² consuming 90.33 mW, as the MLDSA-65 parameter set requires a large scratchpad to simultaneously hold 18 polynomials during the verification process. The ML-KEM encapsulation core is relatively smaller at 2.066 mm² and 53.98 mW. Since both cores never operate at the same time, they share a single Keccak- f [1600] instance, saving approximately 0.166 mm² and bringing the combined total to 6.092 mm² at 144.31 mW.

TABLE II

AREA AND POWER UTILIZATION

Module	Area (mm ²)	Power (mW)
ML-KEM (Encapsulation)	2.066	53.98
ML-DSA (Verification)	4.192	90.33
Keccak sharing saving	~0.166	—
Combined Total	~6.092	~144.31

C. Latency

At 100 MHz, the total authentication latency is approximately 3.35 ms. Shifting in the full 26,567-bit MLDSA-65 signature alongside the 95-bit unlock fields takes around

TABLE III
COMPARISON OF LATENCY WITH PREVIOUS WORK ON IJTAG SECURITY

Work	Authorization Method	Countermeasure	Authorization Latency	Area (mm ²)	PQC
Dworak [3]	n -bit key	Locking SIBs	$< 1 \mu s$	-	No
Rosenfeld [10]	Stream cipher	Trivium stream cipher	$\sim 10 \mu s$	-	No
Kumar [12]	PUF	PUF-LFSR key gating	$\sim 50 \mu s$	-	No
Ithal [13]	SHAKE + Grain-128	TMS hash + output encryption	$\sim 200 \mu s$	-	No
Katti [14]	OTP + FSM	OTP + Trivium + AM	$\sim 1 \mu s$	~ 0.037	No
Proposed Work	ML-KEM + ML-DSA	FIPS 203/204 lattice	$\sim 3.35 \text{ ms}$	~ 6.092	Yes

**Area figures for [3], [10], [12], [13], were not reported in the original works.*

265 μs . Once the frame is captured, the ML-DSA signature verification begins and accounts for the bulk of the latency at 1.5–2.5 ms. The dominant operations here are the ExpandA matrix generation, which requires 30 SHAKE-128 invocations, and the NTT-based matrix-vector multiplication. ML-KEM encapsulation runs once per session and adds approximately 500 μs . The ML-KEM ciphertext CT (8,704 bits) is then shifted out via TDO to the design house, contributing around 87 μs . The OTP-based key comparison, instruction decoding, and data validation are completed in just a few cycles and do not contribute to the overall latency. A comparison of the authentication latency against prior IJTAG security schemes is presented in Table III.

V. CONCLUSION

This work presented an IJTAG security framework built entirely on NIST-standardized post-quantum cryptographic schemes. By integrating ML-KEM for session key establishment and ML-DSA for test command authentication into a centralized Access Manager, the proposed architecture will serve the purpose of enhancing the IJTAG security for post quantum era. This enhancement is primarily a necessity for the devices with long shelf life which are generally used in Automotive, etc. The implementation was done in Verilog and proof of concept simulation is conducted in Cadence Xcelium, and synthesized using the SAED90nm library in Cadence Genus. Both ML-DSA and ML-KEM cores share the same NTT/INTT arithmetic resources. Authentication latency of approximately 3.35 ms is higher than classical alternatives, but this is a trade-off for using lattice-based PQC schemes. The security analysis discussed shows that, the proposed scheme is a countermeasure for scan sniffing and brute force attacks.

ACKNOWLEDGMENT

We would like to acknowledge and thank Abhay G Krishna, Anshul Raghavendra Katti, Akshay C N, and Anekait Thampi for their technical inputs and guidance for this work.

REFERENCES

[1] J. Rearick, B. Eklow, K. Posse, A. Crouch, and B. Bennetts, "IJTAG (Internal JTAG): A step toward a DFT standard," in *Proc. IEEE Int. Conf. Test*, 2005, pp. 8–pp.

[2] F. G. Zadegan, U. Ingelsson, G. Carlsson, and E. Larsson, "Access time analysis for IEEE P1687," *IEEE Trans. Comput.*, vol. 61, no. 10, pp. 1459–1472, 2011.

[3] J. Dworak, A. Crouch, J. Potter, A. Zygmuntowicz, and M. Thornton, "Don't forget to lock your SIB: hiding instruments using P1687," in *Proc. IEEE Int. Test Conf. (ITC)*, 2013, pp. 1–10.

[4] S. Gupta, A. Crouch, J. Dworak, and D. Engels, "Increasing IJTAG bandwidth and managing security through parallel locking SIBs," in *Proc. IEEE Int. Test Conf. (ITC)*, 2017, pp. 1–10.

[5] H. Liu and V. D. Agrawal, "Securing IEEE 1687-2014 standard instrumentation access by LFSR key," in *Proc. IEEE 24th Asian Test Symp. (ATS)*, 2015, pp. 91–96.

[6] A. Zygmuntowicz, J. Dworak, A. Crouch, and J. Potter, "Making it harder to unlock an LSIB: Honeytraps and misdirection in a P1687 network," in *Proc. Design, Autom. Test Europe Conf. Exhib. (DATE)*, 2014, pp. 1–6.

[7] S. Kan, J. Dworak, and J. G. Dunham, "Echeloned IJTAG data protection," in *Proc. IEEE Asian Hardware-Oriented Security Trust (AsianHOST)*, 2016, pp. 1–6.

[8] L. Pierce and S. Tragoudas, "Multi level secure JTAG architecture," in *Proc. IEEE 17th Int. On-Line Testing Symp.*, 2011, pp. 208–209.

[9] C. J. Clark, "Anti-tamper JTAG TAP design enables DRM to JTAG registers and P1687 on-chip instruments," in *Proc. IEEE Int. Symp. Hardware-Oriented Security Trust (HOST)*, 2010, pp. 19–24.

[10] K. Rosenfeld and R. Karri, "Attacks and defenses for JTAG," *IEEE Design Test Comput.*, vol. 27, no. 1, pp. 36–47, 2010.

[11] M. Da Silva, M.-L. Flottes, G. Di Natale, and B. Rouzeyre, "Preventing scan attacks on secure circuits through scan chain encryption," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 38, no. 3, pp. 538–550, 2018.

[12] S. Kumar, N. Satheesh, A. Mahapatra, S. Sahoo, and K. K. Mahapatra, "Securing IEEE 1687 standard on-chip instrumentation access using PUF," in *Proc. IEEE Int. Symp. Nanoelectron. Inf. Syst. (iNIS)*, 2016, pp. 56–61.

[13] N. J. Ithal, N. Adharsh, and K. S. Kumar, "A Complete Security Protocol To Safeguard IJTAG Architecture," in *Proc. IEEE 33rd Asian Test Symp. (ATS)*, 2024, pp. 1–6.

[14] A. R. Katti, A. G. Krishna, A. Thampi, A. C N and S. K. K, "Enforcing Security in IJTAG-Enabled SoCs Through Access Management Mechanism," *2025 IEEE International Symposium on Smart Electronic Systems (iSES)*, Jaipur, India, 2025, pp. 170–175, doi: 10.1109/iSES67504.2025.00041.

[15] National Institute of Standards and Technology, "FIPS 203: Module-Lattice-Based Key Encapsulation Mechanism Standard," Gaithersburg, MD, USA, 2024.

[16] National Institute of Standards and Technology, "FIPS 204: Module-Lattice-Based Digital Signature Standard (ML-DSA)," 2024. [Online]. Available: <https://csrc.nist.gov/pubs/fips/204/final>

[17] M. J. Dworkin, "SHA-3 standard: Permutation-based hash and extendable-output functions," 2015.